COUNTING ARRANGEMENTS: INDISTINGUISHABLE OBJECTS **23/09/25**

1. How many different arrangements are there of the letters:
   a. COOKER
   b. POPPY
   c. BANANA
   d. TARAMASALATA?

2. In how many different orders can colours appear when five balls are drawn without replacement from a bag containing:
   a. one red ball, one yellow, one blue, and two white?
   b. one red ball, one yellow, and three white?
   c. two red balls, and three white?

3. Repeat Question 2, supposing that this time, each ball drawn is immediately replaced.

4. A race at the Olympics has 7 runners. In how many orders can their countries finish if:
   a. there are 3 Americans, 2 Germans, 1 Russian, and 1 Spaniard?
   b. there are 4 Americans, and 3 Germans?

5. How many different six-digit numbers can be made from the numbers:
   a. 3,3,3,4,4,7
   b. 0,0,1,1,1,1 ?

6. How many arrangements of the word TOPOLOGY:
   a. start with a vowel
   b. end with a consonant
   c. start and end with a vowel?

7. Naval signals are made by arranging coloured flags in a vertical line, and the flags are then read from top to bottom. How many signals using six flags can be made if there are:
   a. one green, 3 red, and 2 blue flags?
   b. 2 green, 2 red, and 2 blue flags?
   c. unlimited supplies of green, red, and blue flags?

8. a. In how many arrangements of the letters POODLE are the two Os adjacent?
   b. In how many arrangements of the letters MUGGING are the three Gs adjacent?
   c. In how many arrangements of the letters HERRING are the Rs separated?
   d. [HARD] In how many arrangements of the letters MIGHTIESTTREE are there no instances of the same letter appearing consecutively?

9. How many different arrangements are there of:
   a. four letters from the word HAPPY
   b. three letters from the word MATHEMATICIAN?

10. You have some tiles containing the numbers 0, 0, 0, 0, 0, 1, 2, 2, 3, 3, 3. How many different four-digit even numbers can you create by laying down four of these tiles?

11. The letters of PASSAGES need to be put into the 4 by 2 grid below so that there is least one S in each row. In how many ways can this be done?

12. Six of the letters from WOOLLEN need to be put into a 3 by 2 grid so that that there is at least one vowel in each of the two rows. In how many ways can this be done?